Case Study

Rendi, Ltd.

Rendi, Ltd. was a private company funded under the business umbrella of Otterbox. Based in Trinidad, Colorado it focused on manufacturing of custom designed products. The business utilized a direct marketing model for sales and distribution of product. This began using 3rd party software to manage the multi-tiered payment structure of sales representatives, but because of the large expense I proposed an option to bring the task in house. It was accepted and delivered with great success.

My role was as IT director and lead programmer. As prior attrition diminished the size of our IT staff, we found the production speed and quality raised. This led me to believe in the power of a smaller, focused development team to achieve greater software gains. With this team we developed all aspects of business software for Rendi, Ltd. We used ASP.NET web forms and a custom business layer using MS SQL Server and a custom content management system for our unique business needs.

Our software included a product development engine for art department designers to upload raster art designs and layered transparent images along with meta data that corresponds to a product design and the options of customization allowed for that product. The art UI had permission tools built in for those in the correct position to perform certain duties in the development cycle along with an approval process for design completion.

Once a product was finalized by the art department, sales and marketing added it to the list of hundreds of products stored in the database all with customization attributes equality millions of possibilities for a final product. Sales also utilized the custom UI to manage the large customer base as well as the comp panels for our representative base. This included product sales, and extensive promotion engine, levels of communication to and from representatives as well as their highly complex payment structure.

The customer website was also fully custom with a fully functional shopping cart able to handle personalized products and PCI compliant integrated checkout with multiple forms of payment and billing options. Simple sign-on and Facebook API login and registration options for all user types assisted in the acclaimed user experience.

On the back side was also a fully integrated fulfillment center where the custom CMS system also managed the product to the point of shipping with FedEx, UPS, or USPS. Bar code scanners and individual product tracking was integrated for swift fulfillment handling.

The UI for all aspect of the aforementioned was always intended to be quick and easy. Page actions were to require little thought and instead be more intuitive and reflex oriented to traverse through each section. The server side code was written in Visual Basic with a custom

serialized data layer. REST and SOAP calls were used for many services including ProPay, FedEx, UPS, USPS, Avalara Tax services and many others. Client side code utilized JavaScript/JQuery to assist with user friendly tools as well as page content using AJAX. JQuery was also heavily used for the product designer to have a fluid graphical interface to customize each product. This was optimized to also be mobile friendly and responsive to any screen size. HTML5 and CSS3 were used when possible when backward compatibility could be guaranteed to fail gracefully.
Mark Danielson was the President and Owner of Rendi, Ltd. and was responsible for the day to day operations of the company. This initiative took 1 year to complete.

Case Study Contact
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